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course code : computer vision.

[IOT - Assignment Tool - 1]

(1) how Special resolution : Sampling

how special resolution means the image is sampled with fewer pixels, so small objects lose important details. Increase the Sampling rate resolution by using a higher resolution camera or to capture more pixels.

(2) Dark and noisy night image : Enhancement

use histogram equalization to improve brightness and contrast, along with a noise reduction filter this makes dark regions clearer and unwanted noise.

(3) Continuous Intensity : Quantization

Quantization converts continuous intensity values into a finite number of digital intensity levels. It helps represent the image digitally so that computers can store, process and display it.

(4) Uneven lighting : Image Enhancement

use contrast enhancement such as histogram equalization or adaptive histogram equalization (CLAHE). It improves local contr making defects more visible even in poorly illuminated areas.

(5) Road, vehicles, pedestrains: Segmentation

Image segmentation divides the image into meaningful regions. Such as roads vehicles and Pedestrains. This hulps the autonomous car understand it's surroundings and make safe driving decisions.

(6) Crack detection: Edge detection.

Apply an edge detector such as the Canny edge detection. algorithm to identify sharp intensity changes. cracks appear as edges. allowing the system to locate and inspect defects automatically.

(7) Medical image noise: filtering

use a median filter to remove noise while preserving important edges. This improves the clearer image to make it easier to identify the tumor and surrounding structure

(8) vehicle Identification: object detection

this uses an object detection algorithm. to locate and identify vehicles using boundary boxes. Tracking these vehicles helps monitor traffic flow, congestion, speed and road conditions.

Monitor blur: Image restoration

This blur is mainly caused by object or camera movement during image acquisition. use deblurring / restoration techniques such as motion deconvolution to reduce their blur and recover image details.

Defective vs good products: Image classification.

Image classification assigns such as. each product to categories each product such as based on visual features. it supports automated quality acquisition reducing manual effort and improving inspection, speed and consistency.